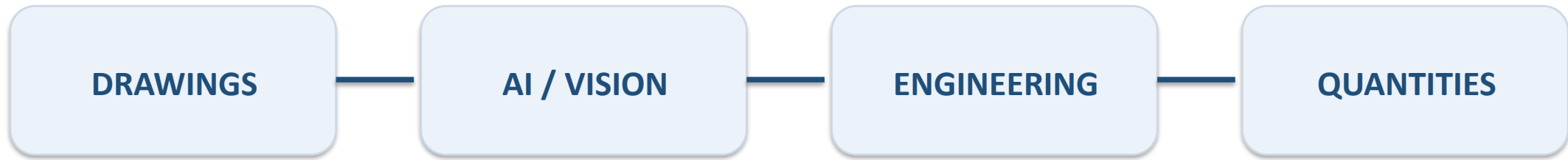


AI-DRIVEN ESTIMATION AUTOMATION



1. Construction Estimation Scope

Major construction materials / work categories identified with the Estimation Team

1. Concrete

HIGH PRIORITY | Current development Focus |
Beam & Slab

2. Steel

HIGH PRIORITY | Current development Focus |
Beam

3. Blockwork

Wall / block quantities

4. Tiles & Granite

Floor / wall finishes

5. Plastering

Internal / external surfaces

6. Painting

Paintable surfaces

7. Waterproofing

Wet areas / terraces

8. Doors & Windows

Openings / glazing

9. Earthwork

Excavation / filling quantities

10. Landscaping

Landscape / external works

11. Plumbing & Electrical

Plumbing / electrical works

12. MS Grill Work

Grill / metal fabrication works

Highlighted focus areas: High Priority - Steel & Concrete

2. Major Structural Components

The structural estimation framework is organized around these core components

1. Beam

Concrete ✓

Steel ✓

Current development focus

2. Slab

Concrete ✓

Current development focus

3. Column

Concrete

Steel

Structural member

4. Foundation

Concrete

Steel

Foundation estimation

5. Shear Wall

Concrete

Steel

Vertical structural wall

6. Retaining Wall

Concrete

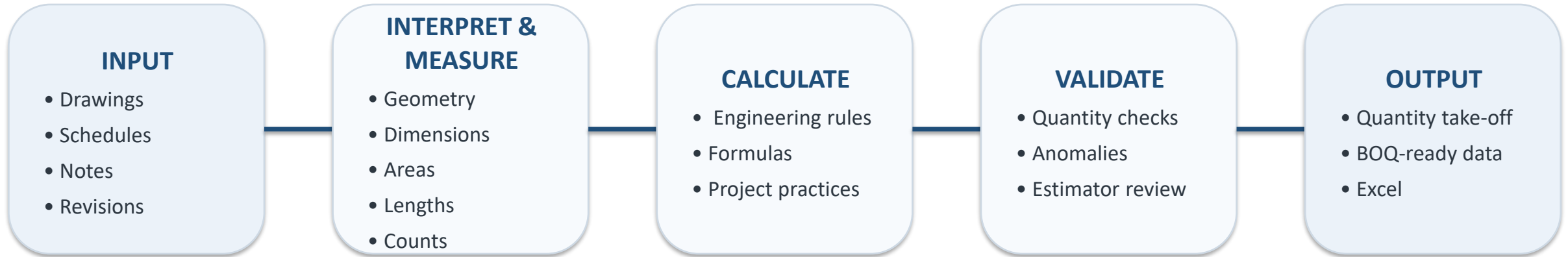
Steel

Earth-support structure

Highlighted focus areas: High Priority - Beam & Slab

3. What Does Construction Estimation Involve?

From engineering information to measurable, reviewable quantities



AI reduces repetitive interpretation effort — engineering logic remains central to quantity calculation.

4. Current AI Development — What Has Been Built

CONCRETE — BEAM & SLAB

- Beam concrete
- Beam shuttering
- Slab concrete
- Slab shuttering
- Deterministic calculation
- LLM assistance for ambiguous inputs

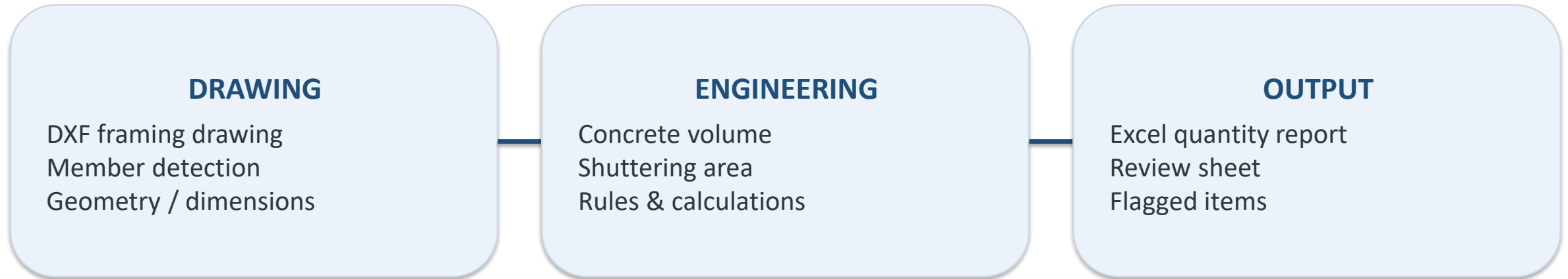
STEEL — BEAM

- Reinforcement drawing understanding
- Beam matching / ownership
- Reinforcement parsing
- Steel quantity + BBS
- Hybrid CAD + Vision
- Deterministic engineering authority

Common foundation: reusable drawing intelligence → material/component-specific estimation engines

5. Concrete Beam & Slab Estimation

Concrete Beam & Slab Estimation Model Link - <http://13.201.15.89>



- Standard drawings can be processed deterministically.
- LLM assistance is used only when information such as slab thickness, tags, beam dimensions or bounded regions is missing / ambiguous.
- Estimation Team validation demonstrated meaningful productivity improvement; quantity accuracy still requires further improvement.

6. Concrete POC - Estimation Team Validation

1

ACCURACY

Current Model Performance

Comparison with manual estimation (Estimation Team)

Work Item	Concrete Accuracy	Shuttering Accuracy
Slab	69%	73%
Beam	78%	81%

2

PRODUCTIVITY

Time Reduction

Reduction in time required for quantity take-off (compared to manual method)

Slab	29% time saving
Beam	67% time saving

3

WHAT THIS MEANS

Key Takeaways

PROCESSING EFFORT REDUCED

Automation delivers clear time savings for quantity take-off.

ACCURACY STILL REQUIRES IMPROVEMENT

Current quantities are not yet accurate enough for direct use.

MANUAL VERIFICATION IS CURRENTLY REQUIRED

Results need to be reviewed and corrected by the Estimation Team.

POC CONCLUSION

Automation demonstrated significant time savings; however, current accuracy requires further development and manual verification before production use.

7. Steel Beam Estimation — Hybrid AI Architecture

AI perception + deterministic engineering calculation



VISION “What reinforcement exists?” → HYBRID RESOLUTION → ENGINEERING “How is it quantified?”

8. Steel Beam Model — Engineering Intelligence Stack

Drawing Architecture

Identity • drawing sets • beam index • versioning

Geometry Intelligence

Regions • sketches • text • leaders • blocks

Detail & Match

Multi-view recognition • context • candidate ranking • confidence

Ownership

Text / leader / bar / stirrup / shared ownership

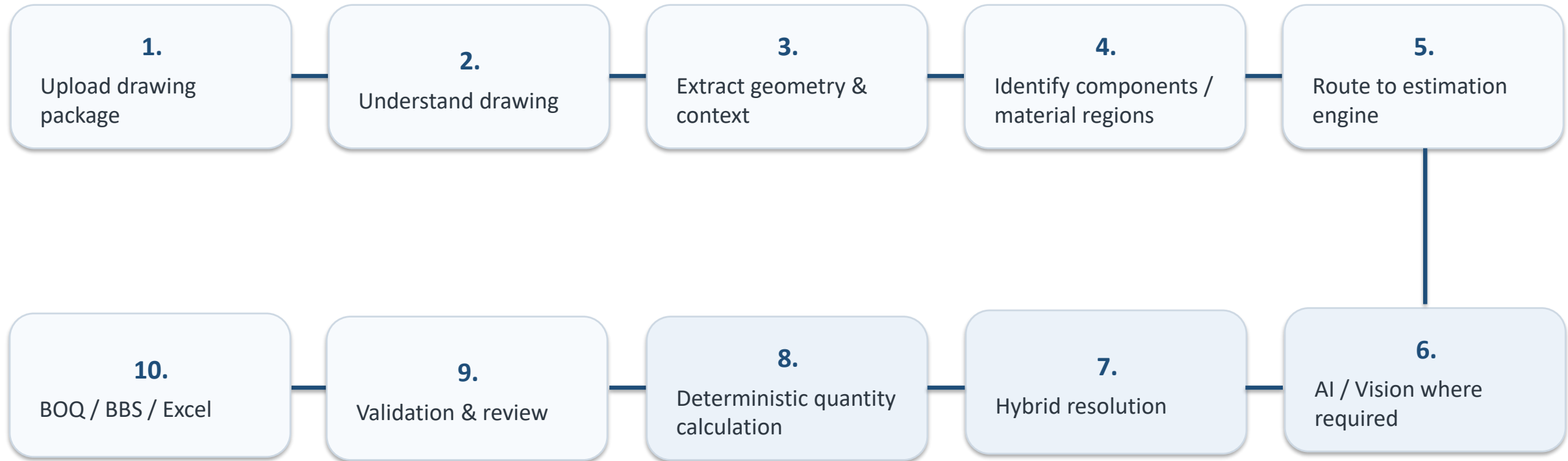
Reinforcement Parsing

Longitudinal bars • stirrups • side-face • laps • hooks • curtailment

Quantity & Output

Cut lengths • unit weights • steel quantity • BBS • Excel

9. End-to-End Steel Estimation Workflow



10. Steel Beam — Internal Accuracy Testing

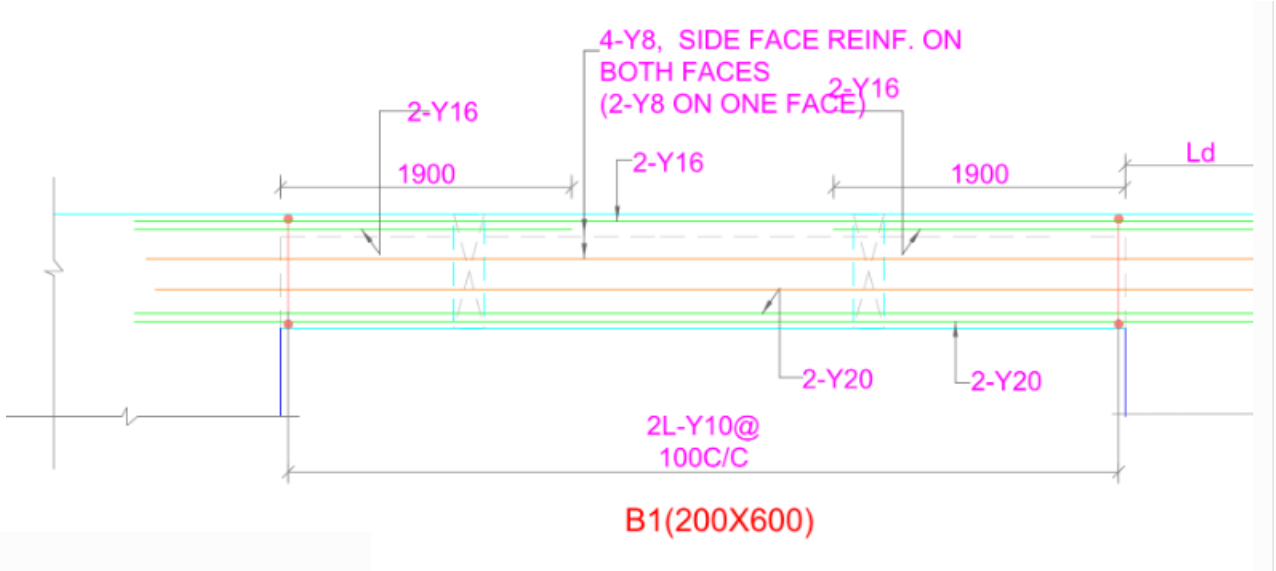
Current model testing result • Subject to Estimation Team validation

Steel Beam Estimation Model Link - <http://13.127.104.99/>

Diameter (mm)	Automated (MT)	Estimated (MT)	Difference	Error Scale	% Error	% Accuracy
Y8	0.55	0.43	0.12	Overestimate	27.57%	72.43%
Y10	0.91	1.18	-0.27	Underestimate	-23.12%	76.88%
Y12	2.31	2.78	-0.47	Underestimate	-16.98%	83.02%
Y16	3.51	2.57	0.94	Overestimate	36.49%	63.51%
Y20	3.00	2.33	0.67	Overestimate	28.54%	71.46%
Y25	1.31	1.12	0.19	Overestimate	17.25%	82.75%
TOTAL	11.58	10.41	1.17	Overestimate	11.24%	88.76%

Internal testing Result: 70-80% Overall Accuracy • Final performance subject to Estimation Team validation.

Cropped Beam Image Sent to Claude



Claude/ Hybrid Answer

B1 — B1(200×600)

B1 detail crop

Claude / Hybrid answer:

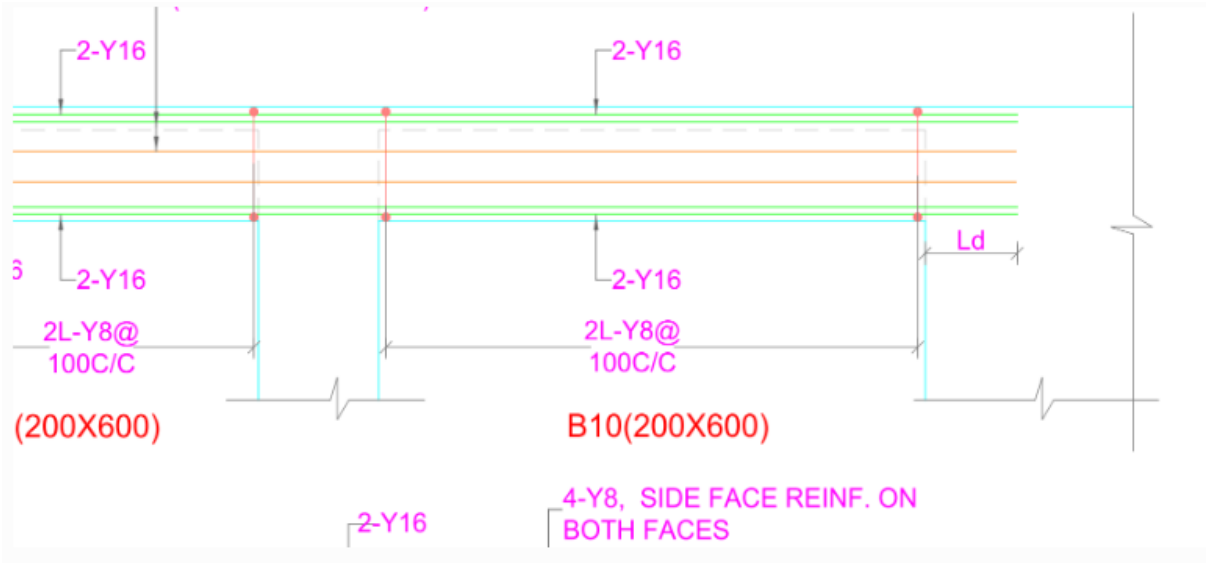
Group	Layer	Spec	Role	Scope
G1	TOP	2-Y16	MAIN	FULL_SPAN
G2	TOP	2-Y16	EXTRA	LEFT_SUPPORT
G3	TOP	2-Y16	EXTRA	RIGHT_SUPPORT
G4	BOTTOM	2-Y20	MAIN	FULL_SPAN
G5	BOTTOM	2-Y20	EXTRA	RIGHT_SUPPORT
G6	SIDE_FACE	4-Y8	EXTRA	FULL_SPAN

Stirrups (Vision): 2L-Y10@100C/C

Estimator Interpretation

SI no		Description	No./Dia.	B(m)/ No.
		SHR & OHT Top		
1.00	TF	B1	1.00	0.20
	TF	Top bars	16.00	2.00
	TF	Top bars - Extra	16.00	2.00
	TF	Bottom bars	20.00	2.00
	TF	Bottom bars	20.00	2.00
	TF	Stirrups	8.00	57.00
	TF	SFR	8.00	4.00

Cropped Beam Image Sent to Claude



Claude/ Hybrid Answer

B10 — B10(200×600)

 B10 detail crop

Claude / Hybrid answer:

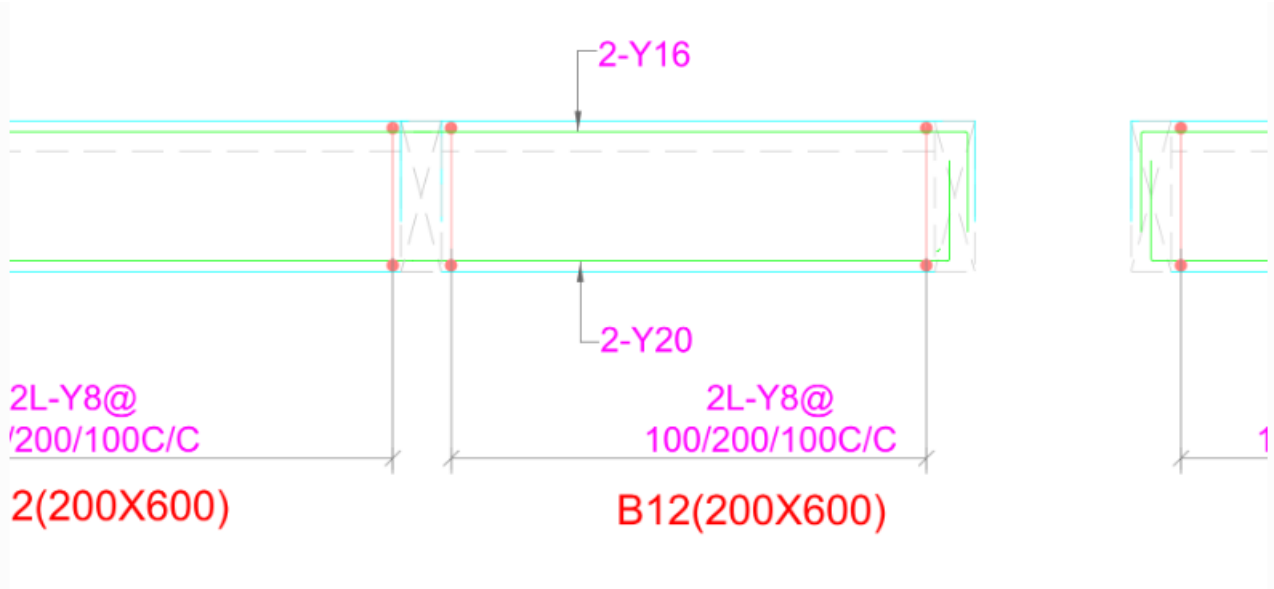
Group	Layer	Spec	Role	Scope
G1	TOP	2-Y16	MAIN	BOTH_SUPPORTS
G2	TOP	2-Y16	EXTRA	LEFT_SUPPORT
G3	TOP	2-Y16	EXTRA	RIGHT_SUPPORT
G4	BOTTOM	2-Y16	MAIN	FULL_SPAN
G5	SIDE_FACE	4-Y8	—	—

Stirrups (Vision): 2L-Y8@100C/C

Estimator Interpretation

10.00	TF	B10	1.00	0.20
	TF	Top bars	16.00	2.00
	TF	Top bars	16.00	2.00
	TF	Bottom bars	16.00	2.00
	TF	Bottom bars	16.00	2.00
	TF	Stirrups	8.00	40.00
	TF	SFR	8.00	4.00

Cropped Beam Image Sent to Claude



Claude/ Hybrid Answer

B12 — B12(200×600)

 B12 detail crop

Claude / Hybrid answer:

Group	Layer	Spec	Role	Scope
G1	TOP	2-Y16	MAIN	FULL_SPAN
G2	BOTTOM	2-Y20	MAIN	FULL_SPAN

Stirrups (Vision): 2L-Y8@100/200/100C/C

Estimator Interpretation

12.00	TF	B12	1.00	0.20
	TF	Top bars	16.00	2.00
	TF	Bottom bars	20.00	2.00
	TF	Stirrups	8.00	45.00
	TF	Stirrups	8.00	11.00

Why Estimation Automation Takes Significant Time

Development effort varies by work item • Engineering understanding + diagnostics + validation

1. Engineering Complexity

Different drawings, structural rules, quantity logic and calculation methods for each work item.

2. Drawing Understanding

Understand how information is represented, connected and interpreted in real construction drawings. This must be transferred to the System.

3. Root-Cause & Diagnostics

Trace incorrect results through multiple stages to identify the actual source of the problem.

4. Solution & Iteration

Implement possible solutions and test them across different drawing conditions and edge cases.

5. Validation & Reliability

Validate outputs against engineering expectations before the automation can be trusted.

Development time = Understanding → Diagnosis → Solution → Iteration → Validation